My doctoral objectives are twofold: First, to theorize how platform workers establish their reliable algorithmic knowledge systems; second, to develop publicly accessible algorithm intervention toolkits by building upon this proposed research. These toolkits are designed to enable a bottom-up, interventionist approach, empowering platform workers to face algorithmic systems more autonomously. My aspiration is to mitigate algorithmic discrimination, labor exploitation, and opacity within platform labor, with the goal of increasing inclusivity and well-being in platform work.

Algorithms play a crucial role in platform work, regulating service and demand and determining evaluation systems. For platform workers, the lack of visibility of these algorithms doesn't mean they're absent from their lives. Instead, it marks the beginning of a challenging journey—akin to an Odyssey—where they must decode these algorithmic 'ghosts' they encounter on the platforms. This necessity to understand and navigate algorithms leads to a kind of folk wisdom that workers rely on to improve their work conditions. Academic research has approached this topic from various perspectives, exploring how workers perceive algorithms (Bucher, 2016; Huang et al., 2022; Karizat et al., 2021), resist (Bonini & Treré, 2024), and manipulate them (Cotter, 2018; Zhang et al., 2022). Shapiro (2017) suggests that as workers become more familiar with platform technologies, they gain autonomy, while traditional expertise becomes less relevant. For instance, taxi drivers are finding digital navigation system more crucial than their own knowledge of the streets (Chen, 2017), highlighting the need for solid algorithmic knowledge to maintain autonomy at work. I contend that this area warrants deeper theoretical exploration. Researchers should delve into the origins, spread, and validation of the algorithmic knowledge that platform workers develop and how it influences their practical use of algorithms on the job. My preference for the term "algorithmic knowledge" over "folk theories" or "algorithmic imaginary" arises because platform workers, unlike casual users, must critically assess their knowledge's reliability, given that their employment status is directly influenced by platform algorithms. When workers find certain pieces of algorithmic knowledge to be practical, they will integrate these strategies into their routines. Conversely, they will abandon them if they're not effective. This approach goes beyond simply understanding algorithms; it alters the workers' actual interaction with the platform and fundamentally affects their job performance.

In my current research, I analyze how dedicated fans decipher and influence social media algorithms, a topic I've presented at Fan Studies Network North America (FSNNA) Annual conference. I interviewed 43 Chinese fan opinion leaders and discovered how key fans decode algorithms and persuade and teach ordinary fans to manipulate these algorithms. This behavior significantly disrupted the recommendation algorithm in China. In response to concerns about fan-driven algorithm manipulation, Weibo launched a separate entertainment trending section in December 2020. However, this measure had a minimal effect, which in turn demonstrates the significant impact and success of fan manipulation of the recommendation algorithm. I describe fans' algorithmic work as an "empirical dance of understanding and manipulating algorithms," since key fans must observe the real-time results and visibility of the algorithmic efforts by ordinary fans, verify the effectiveness of their strategies, and continuously refine them. This project showcases the full process in which a few users decode the algorithm, persuade a large number of other users to participate in algorithmic labor, and ultimately develop a significant algorithmic civil movement. The reason I bring up this project is that it exemplifies how ordinary users disseminate reliable algorithmic knowledge, impart practical and potent algorithmic manipulation practices, foster collective action within a super-community, and intervene in recommendation algorithms. It shows significant potential to develop users' algorithmic practices. In my capacity as a qualitative scholar, I have

extensively researched platform workers' interactions with digital platforms. Several of my papers have been revised and resubmitted to esteemed journals including *Feminist Criminology*, *Feminist Media Studies*, and the *Made in China journal*. Additionally, my work on Chinese fandom and social platforms was featured in the December 2023 issue of *Transformative Works and Cultures*.

Since January 2023, I have collaborated with Professor Y. Connie Yuan from Cornell University on a systematic review of how knowledge is transferred within multinational corporations operating in Africa. This collaboration has enriched my understanding of knowledge creation and transfer processes in management science among other fields, which in turn enhances my research in platform work and algorithmic knowledge studies. These studies on knowledge management explore how knowledge is created and transferred between organizations and nations, examining the conditions that either promote or hinder the creation and transfer of knowledge. Therefore, these literature could help me identify factors that contribute to the varying levels of workers' capacity to acquire, accept, transfer, and apply algorithmic knowledge. For example, many gig economy platforms in various countries originate from other nations, and both cross-cultural and cultural similarity play a role in the transfer of knowledge and learning (Chen et al., 2010). This factor contributes to examining whether individuals can easily understand and accept the algorithmic rules of labor platforms from other countries, and whether they are able to assimilate algorithmic knowledge shared by people from different backgrounds.

Essentially, I am a qualitative researcher with basic skills in Python and other computational areas. During my master's program, I completed the 'Introduction to Python' course at the Oxford Internet Institute, University of Oxford. In 2023, I attended the Oxford Machine Learning Summer School at the University of Oxford, where I engaged in 8 hours of ML fundamentals and 3 hours of ML case studies. While I am open to enhancing my computational skills as required for my PhD, my primary interest lies in social science methodologies, such as interviews, experiments and surveys.

Embarking on a Human-Computer Interaction (HCI) Ph.D. at Carnegie Mellon University, I am poised to cultivate my expertise as a platform and algorithms culture scholar. After completing my Ph.D., I aspire to become a professor while maintaining close collaboration with platform companies to improve the working conditions of platform laborers. In my PhD studies, I am particularly drawn to examining the interplay between algorithmic knowledge and the broader technological frameworks of the platform economy, with an emphasis on how workers navigate and potentially subvert these systems. My ambition is to engage with Dr. Hong Shen to elucidate the opaque aspects of platform algorithms and their contribution to the instability inherent in platform labor, alongside the substantial efforts and challenges digital laborers face in deciphering these algorithms. My objective of systematically explaining the process by which users develop reliable algorithmic knowledge aligns with Dr. Hong Shen's research interests, which include engaging community members in the evaluation and governance of AI and facilitating user intervention in algorithmic processes. I aim to collaborate with Dr. Motahhare Eslami, focusing on empowering users of algorithmic systems to enhance platform workers' autonomy and restore stability through the acquisition of robust algorithmic knowledge. I also plan to seek guidance from Dr. Laura Dabbish on how open data and privacy settings on labor platforms affect the social relationships and information sharing among platform workers. I aspire to gain insights from Dr. Sarah Fox to examine specifically how users, through the creation of their own technological infrastructure, like folk-generated digital products and related tutorials on algorithm resistance, reshape both their identities and actions in the realm of folk algorithmic knowledge.